

c.iii. Fluctuations des distributions de rapidité

On a vu que

$$2\pi \rho_s(\theta) = 1_{[\nu]}^{\text{dr}}(\theta) = \int d\lambda \left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right), \quad (\text{C..18})$$

d'où

$$2\pi \delta\rho_s(\theta) = \int d\lambda C(\theta, \lambda) \rho_s(\lambda) \delta\nu(\lambda). \quad (\text{C..19})$$

Ainsi,

$$\delta\rho(\theta) = \rho_s(\theta) \delta\nu(\theta) + \nu(\theta) \delta\rho_s(\theta) \quad (\text{C..20})$$

$$= \int d\lambda \delta(\theta - \lambda) \rho_s(\lambda) \delta\nu(\lambda) + \int d\lambda \nu(\theta) C(\theta, \lambda) \rho_s(\lambda) \delta\nu(\lambda), \quad (\text{C..21})$$

$$= \int d\lambda \left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right) \rho_s(\lambda) \delta\nu(\lambda). \quad (\text{C..22})$$

En réinjectant (C..11) et (C..22) dans la définition

$$\langle \delta\rho(\theta') \delta\rho(\theta) \rangle_w = - \left[L \frac{\delta^2(\mathcal{S}_{YY} - \mathcal{W})}{\delta\rho \delta\rho} \right]^{-1}(\theta, \theta'), \quad (\text{C..23})$$

on obtient finalement

$$\langle \delta\rho(\theta') \delta\rho(\theta) \rangle_w = -\frac{1}{L} \int d\lambda \frac{\left(\delta(\theta - \lambda) + \nu(\theta) C(\theta, \lambda) \right) \left(\delta(\theta' - \lambda) + \nu(\theta') C(\theta', \lambda) \right)}{s''(\nu(\lambda))} \rho_s(\lambda) \quad (\text{C..24})$$

$$= \frac{1}{L} \mathcal{D}^{-1}(\theta, \theta') + \frac{1}{L} \mathcal{B}(\theta, \theta'), \quad (\text{C..25})$$

où

$$\mathcal{D}^{-1}(\theta, \theta') = (\rho_s(\theta) \nu(\theta) (1 - \nu(\theta))) \delta(\theta - \theta'), \quad (\text{C..26})$$

$$\mathcal{B}(\theta, \theta') = \left[\nu(\theta') \rho_s(\theta) \nu(\theta) (1 - \nu(\theta)) + \nu(\theta) \rho_s(\theta') \nu(\theta') (1 - \nu(\theta')) \right] C(\theta, \theta') \quad (\text{C..27})$$

$$+ \nu(\theta) \nu(\theta') \int d\lambda \rho_s(\lambda) \nu(\lambda) (1 - \nu(\lambda)) C(\theta, \lambda) C(\theta', \lambda). \quad (\text{C..28})$$

Cette expression coïncide numériquement avec le résultat obtenu en (IV.60).

Bibliographie

- [DS17] Benjamin DOYON et Herbert SPOHN. “Drude Weight for the Lieb-Liniger Bose Gas”. In : *SciPost Physics* 3.6 (déc. 2017). ISSN : 2542-4653. DOI : [10.21468/scipostphys.3.6.039](https://doi.org/10.21468/scipostphys.3.6.039). URL : <http://dx.doi.org/10.21468/SciPostPhys.3.6.039>.

